Rational Matrix Policy Optimization for Rational Feed-Forward Language Models

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Abstract

1 Introduction

2 Background and Related Work

3 Rational Matrix Policy Optimization

Let $\mathcal{L}(\theta)$ be the language-model training objective. For layer l, the RLB block projects the residual-stream input $x_l \in \mathbb{R}^d$ into G groups of width m,

$$z_l = A_l x_l, \qquad z_{l,g} = \operatorname{group}_g(z_l), \tag{1}$$

normalizes each group by its RMS radius,

$$r_{l,g} = \sqrt{m^{-1}\|z_{l,g}\|_2^2 + \epsilon_{\text{rms}}}, \qquad u_{l,g} = z_{l,g}/r_{l,g}, \tag{2}$$

and applies a learned rational map in normalized coordinates,

$$h_{l,g} = r_{l,g} R_{l,g}(u_{l,g}), \qquad y_l = B_l \operatorname{concat}_{g=1}^{G} h_{l,g}. \tag{3}$$

The input matrix A_l selects rational domains, the coefficients of $R_{l,g}$ shape the group response, and the output matrix B_l recombines rational features. MatrixPolicy uses this separation by partitioning parameters as

$$\theta = \theta_{\text{base}} \cup \theta_{\text{coeff}} \cup \{A_l, B_l\}_{l=1}^{L}. \tag{4}$$

The base parameters use the same optimizer as the non-rational baseline. Rational coefficients use a coefficient rule. MatrixPolicy is applied only to A_l and B_l.

In the homogeneous case $\epsilon_{\text{rms}} = 0$, and for nonzero group radii, each group has a positive scale gauge:

$$A_{l,g} \mapsto a_g A_{l,g}, \qquad B_{l,g} \mapsto a_g^{-1} B_{l,g}, \qquad a_g > 0. \tag{5}$$

The represented function is unchanged because $z_{l,g}$ and $r_{l,g}$ scale together, $u_{l,g}$ is unchanged, and the inverse scale in B_l cancels the scale in $h_{l,g}$. With a nonzero RMS floor, the exact quotient dynamics are replaced by the residual model in Appendix B. This matters because optimizer state is stored in coordinates, not in the quotient space of represented functions.

At step t, each matrix role $r \in \{\text{in}, \text{out}\}$ receives a role-depth factor $\rho_r(l)$ and an on-policy statistic factor $s_{\text{stat}}(l, r, t)$. The matrix update is

$$\begin{aligned} M_{l,r}^{t+1} = M_{l,r}^{t} &+ \Delta_{\text{adapt}}(M_{l,r}; \eta_t a_{\text{mat}}(l,r,t)[1-\mu(l,r,t)]) \\ &+ \Delta_{\text{mat}}(M_{l,r}; \eta_t a_{\text{matnorm}} \mu(l,r,t)), \end{aligned} \tag{6}$$

where $M_{l,\text{in}} = A_l$, $M_{l,\text{out}} = B_l$, η_t is the shared base learning rate, Δ_{adapt} is an Adam-style adaptive step, and Δ_{mat} is a short early matrix-normalized step. The mixture coefficient

$\mu(l,r,t)$ is active only early in training. After the matrix update, MatrixPolicy computes a target log-ratio for each group and applies the bounded rebalance

$$A_{l,g} \leftarrow e^{\ell_g} A_{l,g}, \qquad B_{l,g} \leftarrow e^{-\ell_g} B_{l,g}. \tag{7}$$

The update changes the representative used by the optimizer while preserving, or with $\epsilon_{\text{rms}} > 0$ approximately preserving, the represented block function. Full hyperparameter definitions are in Appendix A.

Algorithm 1: Rational Matrix Policy step. Given gradients $\nabla\mathcal{L}(\theta^t)$ and base learning rate η_t:

1. Update exponential moving averages of rational-group pressures.
2. Compute role-depth and group-stat multipliers for each RLB matrix.
3. Update non-rational parameters with the base optimizer and update rational coefficients with the coefficient rule.
4. Update each A_l and B_l with the MatrixPolicy mixture step.
5. Apply the bounded positive-gauge rebalance to each rational group.

4 Experiments

5 Discussion

6 Conclusion

A MatrixPolicy Details

Role-depth factors. For layers indexed by $l = 1, \ldots, L$ with $L > 1$, define normalized layer depth $d_l = (l-1)/(L-1)$ and set

$$\rho_{\text{in}}(l) = \text{clip}(1 - 0.50(d_l - 0.5), 0.55, 1.40), \tag{8}$$
$$\rho_{\text{out}}(l) = \text{clip}(1 + 1.00(d_l - 0.5), 0.55, 1.40), \tag{9}$$
$$a_{\text{role}}(l,r) = \max(0.10, 1 + 1.20(\rho_r(l) - 1)). \tag{10}$$

Let $\xi_t = t/T$ be training progress and let $\sigma_{a,b}(\xi)$ denote the smoothstep gate from 0 to 1 over $[a,b]$. The statistic multiplier $a_{\text{stat}}(l,r,t) > 0$ is the layer/role aggregate of the group preconditioners defined below, normalized to one when group statistics are not active. The local adaptive scale is

$$a_{\text{mat}}(l,r,t) = \text{clip}(3.0\, a_{\text{role}}(l,r) a_{\text{stat}}(l,r,t), 0.40, 4.0). \tag{11}$$

The matrix-normalized branch uses multiplier $a_{\text{matnorm}} > 0$ and mixture weight $\mu(l,r,t) \in [0,1]$, a smoothstep-gated function of ξ_t, layer depth, matrix role, and $a_{\text{stat}}(l,r,t)$.

On-policy group statistics. All ratios and logarithms below are evaluated on strictly positive stabilized RMS, EMA, and gain quantities; the displayed formulas omit the corresponding epsilon floors. For group g, MatrixPolicy tracks relative gradient pressures

$$p_{\text{in},g} = \frac{\text{rms}(\nabla A_{l,g})}{\text{rms}(A_{l,g})}, \qquad p_{\text{out},g} = \frac{\text{rms}(\nabla B_{l,g})}{\text{rms}(B_{l,g})}, \qquad p_{\text{rat},g} = \text{rms}(\nabla \theta_{\text{coeff},g}). \tag{12}$$

With EMAs $q_{\text{in},g}$, $q_{\text{out},g}$, and $q_{\text{rat},g}$, we define

$$\pi_g = \log q_{\text{in},g} - \log q_{\text{out},g}, \qquad \alpha_g = \log q_{\text{rat},g} - \frac{1}{2}(\log q_{\text{in},g} + \log q_{\text{out},g}). \tag{13}$$

Group-stat preconditioning. The group preconditioner is centered and clipped:

$$c_g = c_{\text{gain},g} c_{\text{pressure},g} c_{\text{activity},g}, \qquad c_g \leftarrow c_g / \operatorname{geomean}(c), \qquad c_g \leftarrow \operatorname{clip}(c_g, 0.75, 1.35). \tag{14}$$

Here $k_g > 0$ is the role-specific group gain used for matrix preconditioning: the derivative-RMS gain for A_l and the output-RMS gain for B_l. The factors are

$$c_{\text{gain},g} = \left(\frac{\operatorname{geomean}(k)}{k_g} \right)^{0.20}, \qquad c_{\text{pressure},g} = \exp(0.10\, d_{\text{pressure},g}), \tag{15}$$

where $d_{\text{pressure},g} = -\pi_g$ for A_l and $+\pi_g$ for B_l. The activity damping term is

$$c_{\text{activity},g} = \exp[-0.20 \ \operatorname{ReLU}((\alpha_g - 0.05)/0.45)]. \tag{16}$$

Gauge rebalance. Let

$$\gamma_g = \log \operatorname{rms}(A_{l,g}) - \log \operatorname{rms}(B_{l,g}). \tag{17}$$

The target log-ratio is

$$\tau_g = \frac{1}{2}(\log d_g^{\text{gain}} - \log o_g^{\text{gain}}) + \beta_{\text{pressure}}(\log q_{\text{in},g} - \log q_{\text{out},g}). \tag{18}$$

Here $d_g^{\text{gain}} > 0$ and $o_g^{\text{gain}} > 0$ are the derivative and output group-gain estimates, and β_{pressure} is the pressure balance coefficient. Let $s(t,l) \in [0,1]$ be the rebalance schedule, $a_g^{\text{activity}} \in (0,1]$ the activity damping multiplier, and $\ell_{\max} > 0$ the maximum absolute log-scale move. The exact target step is $\ell_g^\star = \frac{1}{2}(\tau_g - \gamma_g)$, while the applied bounded move is

$$\ell_g = \operatorname{clip}(s(t,l) a_g^{\text{activity}} \ell_g^\star, -\ell_{\max}, \ell_{\max}). \tag{19}$$

The clipped update maps γ_g to $\gamma_g + 2\ell_g$, so it is a bounded move toward τ_g rather than exact canonicalization unless clipping and schedule damping are inactive. The transform $A_{l,g} \leftarrow e^{\ell_g} A_{l,g}$ and $B_{l,g} \leftarrow e^{-\ell_g} B_{l,g}$ preserves the homogeneous RLB function exactly under the nonzero-radius condition; the stabilized implementation is covered by the residual perturbation model in Appendix B.

B A Conditional Quotient-Optimization Separation

B.1 Quotient Geometry

For one RLB layer, suppress the layer index and write

$$z_g = A_g x, \tag{20}$$

$$r_g = \sqrt{m^{-1} \|z_g\|_2^2 + \epsilon_{\text{rms}}}, \tag{21}$$

$$u_g = z_g / r_g, \tag{22}$$

$$h_g = r_g R_g(u_g), \tag{23}$$

$$y = B \operatorname{concat}_{g=1}^{G} h_g. \tag{24}$$

For $a_g > 0$, define

$$T_a(A_g, B_g) = (a_g A_g, a_g^{-1} B_g), \qquad g = 1, \ldots, G. \tag{25}$$

Equivalently, with $s_g = \log a_g$,

$$T_s(A_g, B_g) = (e^{s_g} A_g, e^{-s_g} B_g). \tag{26}$$

Let $\mathcal{P}$ be the local RLB-matrix parameter space, let $\mathcal{G} = (\mathbb{R}, +)^G$, and let $\mathcal{Q} = \mathcal{P}/\mathcal{G}$.

Lemma B.1 (Exact positive gauge). If $\epsilon_{\text{rms}} = 0$ and $r_g > 0$ for every group on the considered input, then $T_s(A, B)$ and (A, B) represent the same layer map for all $s \in \mathbb{R}^G$.

Proof. For one group, $A'_g = e^{s_g} A_g$ gives $z'_g = e^{s_g} z_g$. Since $\epsilon_{\text{rms}} = 0$, $r'_g = e^{s_g} r_g$, hence $u'_g = u_g$ and $h'_g = e^{s_g} h_g$. The corresponding output block satisfies $B'_g h'_g = e^{-s_g} B_g e^{s_g} h_g = B_g h_g$. Summing over groups gives the claim. $\square$

Assumption B.1 (Local quotient chart). In a neighborhood of the trajectory, parameters can be written as

$$\theta = \theta(q, s), \qquad q \in \mathbb{R}^d, \qquad s \in \mathbb{R}^G, \tag{27}$$

where q is a local coordinate on $\mathcal{Q}$ and s is the log-gauge coordinate. The objective satisfies

$$L(\theta(q, s)) = F(q). \tag{28}$$

For the exact statements, $\epsilon_{\text{rms}} = 0$. Nonzero RMS stabilization is included only through the residual model of Assumption B.5.

Assumption B.2 (Local quadratic model). The quotient objective is

$$F(q) = F_\star + \frac{1}{2}(q - q_\star)^\top H(q - q_\star), \tag{29}$$

where $H \succ 0$ and $0 < \mu I \preceq H \preceq LI < \infty$.

Lemma B.2 (Gauge tangent orthogonality). Under Lemma B.1, for any differentiable loss depending on the RLB matrices only through the represented function,

$$\langle \nabla_{A_g} L, A_g \rangle - \langle \nabla_{B_g} L, B_g \rangle = 0. \tag{30}$$

Proof. Differentiate $L(e^{s_g} A_g, e^{-s_g} B_g) = L(A_g, B_g)$ at $s_g = 0$. $\square$

B.2 Canonicalized Quotient Preconditioning

For group g, define

$$\gamma_g = \log \operatorname{rms}(A_g) - \log \operatorname{rms}(B_g). \tag{31}$$

The logarithms in this subsection are taken on true positive RMS quantities. Stabilized RMS floors preserve the lemma exactly only when inactive; otherwise the resulting discrepancy is part of the perturbation model in Assumption B.5. Given a target log-ratio τ_g, the unclipped rebalance is

$$\ell_g = \frac{1}{2}(\tau_g - \gamma_g), \qquad A_g^+ = e^{\ell_g} A_g, \qquad B_g^+ = e^{-\ell_g} B_g. \tag{32}$$

Lemma B.3 (RMS-ratio canonicalization). For true positive RMS values, the update in equation 32 satisfies

$$\log \operatorname{rms}(A_g^+) - \log \operatorname{rms}(B_g^+) = \tau_g. \tag{33}$$

Proof. By homogeneity of RMS under positive scaling,

$$\log \operatorname{rms}(A_g^+) - \log \operatorname{rms}(B_g^+) = \log \operatorname{rms}(e^{\ell_g} A_g) - \log \operatorname{rms}(e^{-\ell_g} B_g) \tag{34}$$

$$= \gamma_g + 2\ell_g = \tau_g. \tag{35}$$

$\square$

Lemma B.4 (Gauge-dependent raw quotient step). Let $w > 0$, $a = e^s \sqrt{w}$, $b = e^{-s}\sqrt{w}$, and $\ell(w) = \frac{1}{2}(w - y)^2$. One Euclidean gradient-descent step in (a, b) with step size η gives

$$w^+ = w - 2\eta w \cosh(2s)(w - y) + O(\eta^2). \tag{36}$$

Proof. Since $\partial \ell / \partial a = (w - y)b$ and $\partial \ell / \partial b = (w - y)a$,

$$w^+ = (a - \eta(w - y)b)(b - \eta(w - y)a) \tag{37}$$

$$= w - \eta(w - y)(a^2 + b^2) + O(\eta^2). \tag{38}$$

Using $a^2 + b^2 = w(e^{2s} + e^{-2s}) = 2w \cosh(2s)$ gives equation 36. $\square$

Assumption B.3 (Frozen quotient preconditioners). During the local phase, MatrixPolicy and AdamW induce fixed positive definite quotient preconditioners P_{MP} and P_{AdamW} satisfying

$$q_{t+1}^{\mathrm{MP}} = q_t^{\mathrm{MP}} - \eta_{\mathrm{MP}} P_{\mathrm{MP}} \nabla F(q_t^{\mathrm{MP}}), \tag{39}$$

$$q_{t+1}^{\mathrm{AdamW}} = q_t^{\mathrm{AdamW}} - \eta_{\mathrm{AdamW}} P_{\mathrm{AdamW}} \nabla F(q_t^{\mathrm{AdamW}}). \tag{40}$$

Here P_{MP} is the quotient preconditioner after gauge rebalance and P_{AdamW} is the quotient preconditioner induced by AdamW's raw-coordinate adaptive state.

For $P \succ 0$, define

$$K(P) = P^{1/2} H P^{1/2}, \qquad \kappa(P; H) = \frac{\lambda_{\max}(K(P))}{\lambda_{\min}(K(P))}. \tag{41}$$

Let

$$\eta^\star(P) = \frac{2}{\lambda_{\max}(K(P)) + \lambda_{\min}(K(P))}, \qquad \rho(P; H) = \frac{\kappa(P; H) - 1}{\kappa(P; H) + 1}. \tag{42}$$

Assumption B.4 (Strict quotient-conditioning advantage).

$$\kappa_{\mathrm{MP}} := \kappa(P_{\mathrm{MP}}; H) < \kappa(P_{\mathrm{AdamW}}; H) =: \kappa_{\mathrm{AdamW}}. \tag{43}$$

B.3 Contraction and Perturbation Bounds

Lemma B.5 (Preconditioned quadratic contraction). Let F satisfy Assumption B.2. For $P \succ 0$, consider $q_{t+1} = q_t - \eta^\star(P) P \nabla F(q_t)$. Then

$$F(q_T) - F_\star \le \rho(P; H)^{2T} \big(F(q_0) - F_\star\big), \tag{44}$$

and the bound is sharp.

Proof. Let $e_t = q_t - q_\star$ and $\widetilde{e}_t = P^{-1/2} e_t$. Then

$$\widetilde{e}_{t+1} = (I - \eta^\star(P) K(P))\, \widetilde{e}_t, \qquad 2(F(q_t) - F_\star) = \widetilde{e}_t^T K(P) \widetilde{e}_t. \tag{45}$$

Diagonalize $K(P) = U \Lambda U^T$ and set $v_t = U^T \widetilde{e}_t$. The largest multiplier satisfies

$$\max_i |1 - \eta^\star(P)\lambda_i| = \frac{\lambda_{\max}(K(P)) - \lambda_{\min}(K(P))}{\lambda_{\max}(K(P)) + \lambda_{\min}(K(P))} = \rho(P; H). \tag{46}$$

Thus

$$2(F(q_{t+1}) - F_\star) = \sum_i \lambda_i v_{t+1,i}^2 \tag{47}$$

$$\le \rho(P; H)^2 \sum_i \lambda_i v_{t,i}^2 \tag{48}$$

$$= 2\rho(P; H)^2 (F(q_t) - F_\star). \tag{49}$$

Iteration gives equation 44. Equality holds when the transformed initial error $\widetilde{e}_0 = P^{-1/2}(q_0 - q_\star)$ is an eigenvector of $K(P)$ associated with $\lambda_{\min}(K(P))$ or $\lambda_{\max}(K(P))$. □

Theorem B.1 (Conditional optimization separation). Suppose Assumptions B.1–B.4 hold, and set $\eta_{\mathrm{MP}} = \eta^\star(P_{\mathrm{MP}})$ and $\eta_{\mathrm{AdamW}} = \eta^\star(P_{\mathrm{AdamW}})$. Then, for every integer $T \ge 1$,

$$\sup_{q_0 \ne q_\star} \frac{F(q_T^{\mathrm{MP}}) - F_\star}{F(q_0) - F_\star} = \left(\frac{\kappa_{\mathrm{MP}} - 1}{\kappa_{\mathrm{MP}} + 1}\right)^{2T} < \left(\frac{\kappa_{\mathrm{AdamW}} - 1}{\kappa_{\mathrm{AdamW}} + 1}\right)^{2T} = \sup_{q_0 \ne q_\star} \frac{F(q_T^{\mathrm{AdamW}}) - F_\star}{F(q_0) - F_\star}. \tag{50}$$

Proof. The map $\kappa \mapsto (\kappa - 1)/(\kappa + 1)$ is strictly increasing for $\kappa > 0$. Assumption B.4 gives $\rho(P_{\mathrm{MP}}; H) < \rho(P_{\mathrm{AdamW}}; H)$. Lemma B.5 applied to P_{MP} and P_{AdamW} gives the displayed factors, and sharpness gives both equalities. □

Assumption B.5 (Uniform local perturbation bound). For a fixed horizon T, the implemented quotient iterates satisfy

$$q_{t+1}^{\mathrm{MP,impl}} = q_t^{\mathrm{MP,impl}} - \eta_{\mathrm{MP}} P_{\mathrm{MP}} \nabla F(q_t^{\mathrm{MP,impl}}) + r_t^{\mathrm{MP}}, \tag{51}$$

$$q_{t+1}^{\mathrm{AdamW,impl}} = q_t^{\mathrm{AdamW,impl}} - \eta_{\mathrm{AdamW}} P_{\mathrm{AdamW}} \nabla F(q_t^{\mathrm{AdamW,impl}}) + r_t^{\mathrm{AdamW}}, \tag{52}$$

with $\left\|r_t^{\mathrm{MP}}\right\|_2 + \left\|r_t^{\mathrm{AdamW}}\right\|_2 \leq \delta$ for all $0 \leq t < T$. The residuals include the RMS floor, denominator floors, clipping, moment lag, weight decay, and nonquadratic Taylor remainders.

Proposition B.1 (Stability of a strict finite-time gap). Fix $T \geq 1$ and suppose the idealized dynamics satisfy

$$m_T := F(q_T^{\mathrm{AdamW}}) - F(q_T^{\mathrm{MP}}) > 0. \tag{53}$$

Assume the implemented and ideal recursions use the same initial points, and assume the ideal trajectories lie in a compact set K with a compact neighborhood U on which F and the ideal one-step maps are Lipschitz. Then there exists $\delta_0 > 0$ such that, whenever Assumption B.5 holds with $\delta < \delta_0$, the implemented trajectories remain in U through time T and

$$F(q_T^{\mathrm{AdamW,impl}}) - F(q_T^{\mathrm{MP,impl}}) > \frac{m_T}{2} > 0. \tag{54}$$

Proof. Let Ψ_{MP} and Ψ_{AdamW} be the ideal one-step maps, and let C_Ψ be a common Lipschitz constant on U. Let $r_{\mathrm{stay}} > 0$ be the distance from the finite ideal trajectories to U^c. By induction, as long as the implemented trajectories remain in U,

$$\left\|q_t^{\mathrm{MP,impl}} - q_t^{\mathrm{MP}}\right\| + \left\|q_t^{\mathrm{AdamW,impl}} - q_t^{\mathrm{AdamW}}\right\| \leq C_T \delta \tag{55}$$

for a constant C_T depending only on T and C_Ψ. Taking $\delta \leq r_{\mathrm{stay}}/(2C_T)$ keeps the implemented trajectories in U and closes the induction. If C_F is a Lipschitz constant for F on U, then

$$\left|F(q_T^{\mathrm{MP,impl}}) - F(q_T^{\mathrm{MP}})\right| + \left|F(q_T^{\mathrm{AdamW,impl}}) - F(q_T^{\mathrm{AdamW}})\right| \leq C_F C_T \delta. \tag{56}$$

Taking

$$\delta_0 = \min\left\{\frac{m_T}{2C_F C_T}, \frac{r_{\mathrm{stay}}}{2C_T}\right\} \tag{57}$$

gives the claim. □